SOLPS5.0: Atomic physics in B2.5

David Coster, ...

- in which we cover the atomic physics used in B2.5

Choice of source for atomic physics data

- The main file used to specify the atomic physics in B2.5 is `b2ar.dat`
- the main choice is ADPAK, STRAHL or ADAS

  ADPAK

  STRAHL

  ADAS
- our current recommendation is to use ADAS
- for hydrogen, the ionization, recombination and cooling rate data for all three options is overwritten by data from Weisheit
  - but not the CX data
- the original STRAHL database had no data for CX rates
  - so Bas added a formula
    * that should be OK for H/D/T
    * but can be wildly wrong for other species
- ADAS has CX data only for C
  - not even for H/D/T
  - new option added to allow Bas' fit formula to be used

b2ar.dat

```
*tlohi             (tlo, thi; real, free format)
  1.0e-1  1.0e4
*nlohi             (nlo, nhi; real, free format)
  1.0e18  1.0e22
*numnuc            (nnuc; integer, free format)
   3
*nucspec           (nz, izlo, izhi; integers, one triple per line)
   1  0  1
   6  0  6
   2  0  2
*flag              (either 'adpak', 'strahl' or 'adas', free format)
  'adas'
*tailep            (real, free format)
  0.08
```

tlohi range of temperatures for atomic physics table

nlohi range of densities for atomic physics table

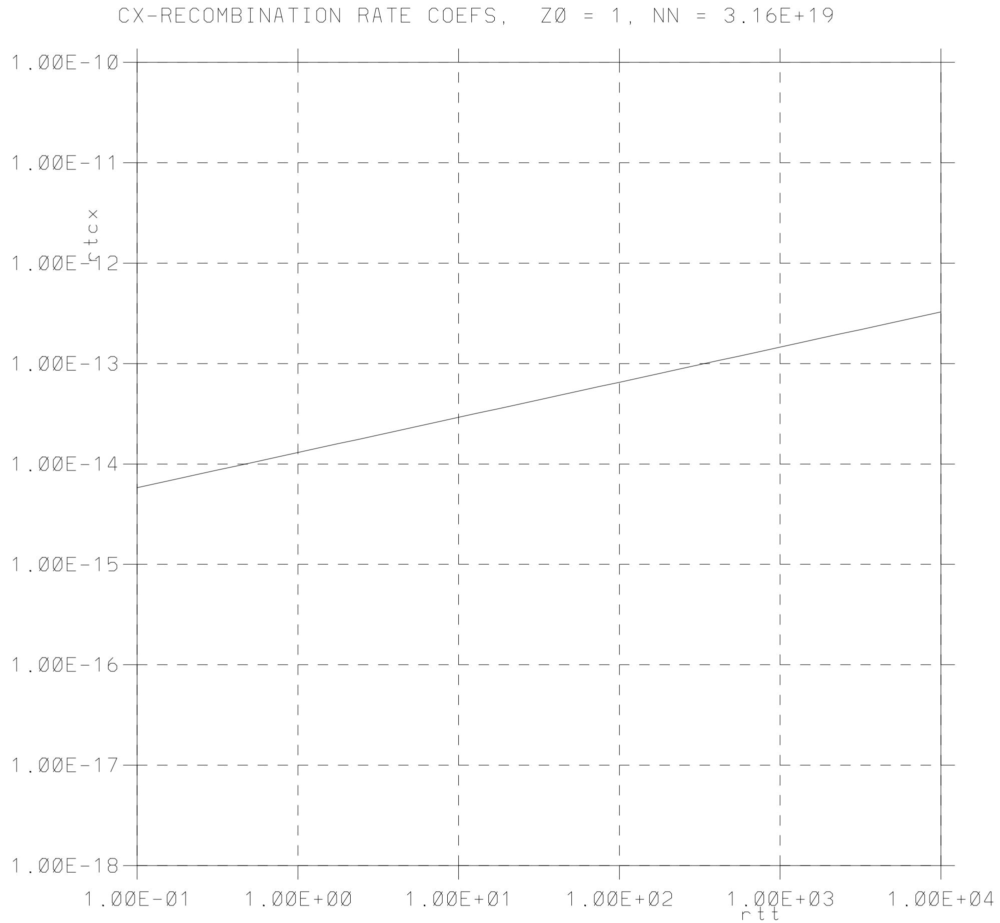
numnuc number of distinct species

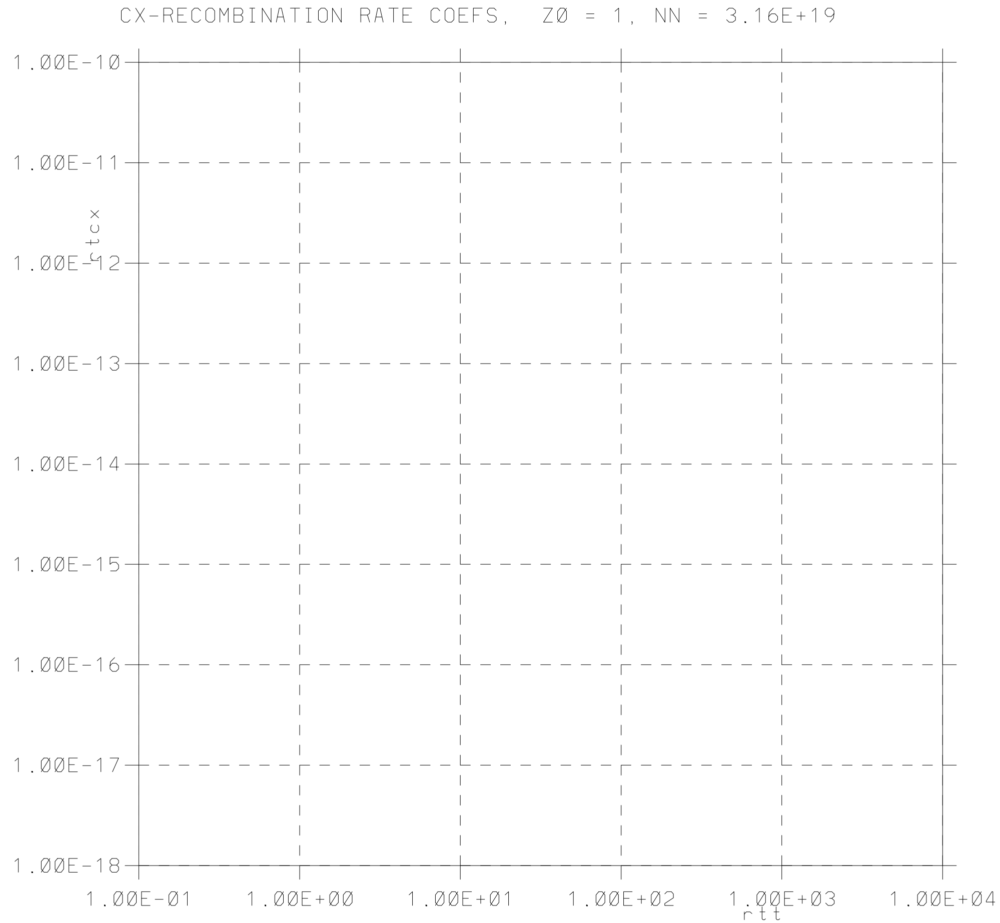
nucspec nuclear charge, lowest charge state, highest charge state (here for H, C & He)

flag which atomic physics database to use

tailep flag used for extrapolation ???

STRAHL versus ADAS rates: H CX

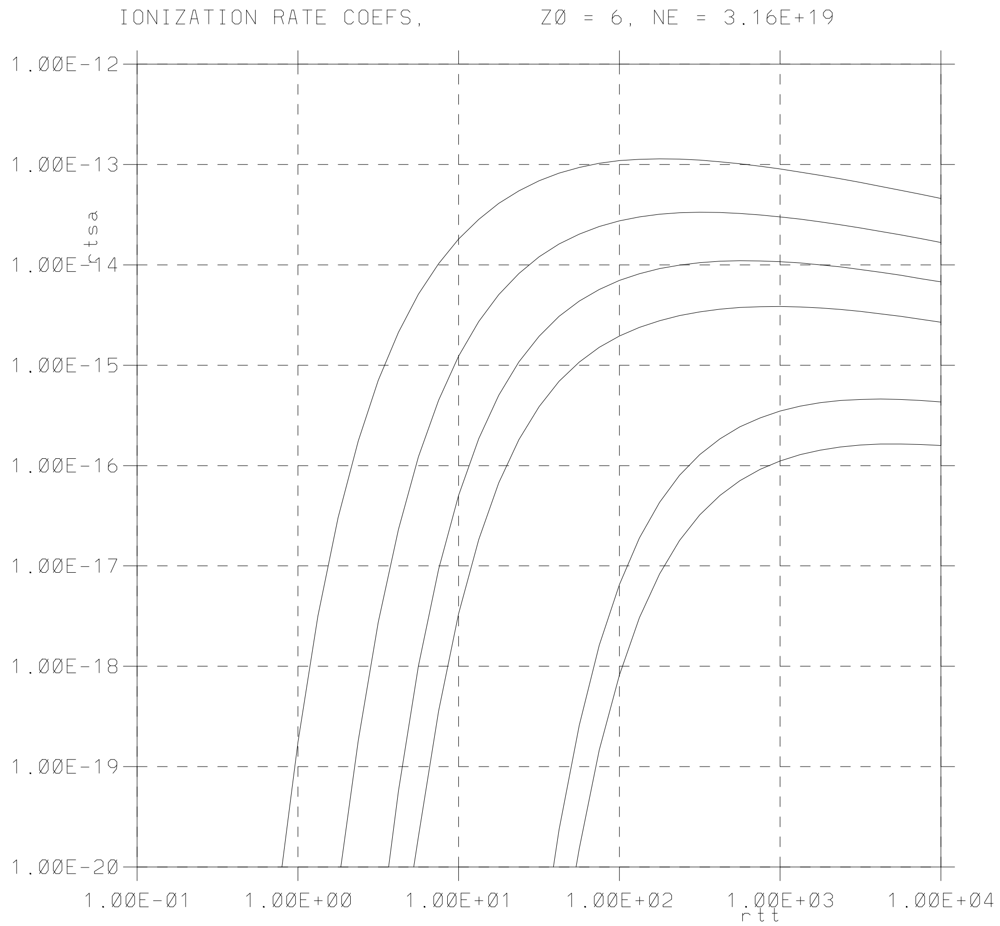




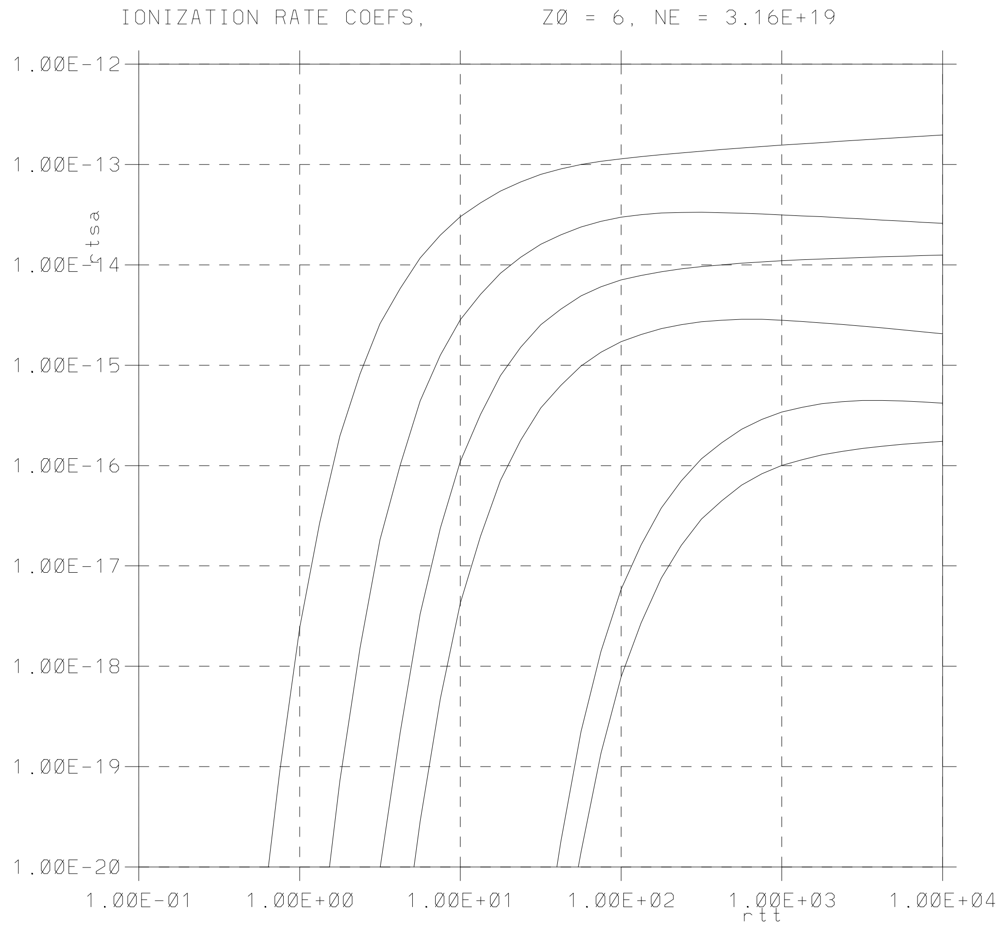
No ADAS H CX data!

STRAHL versus ADAS rates: differences in the rates for C

STRAHL



ADAS

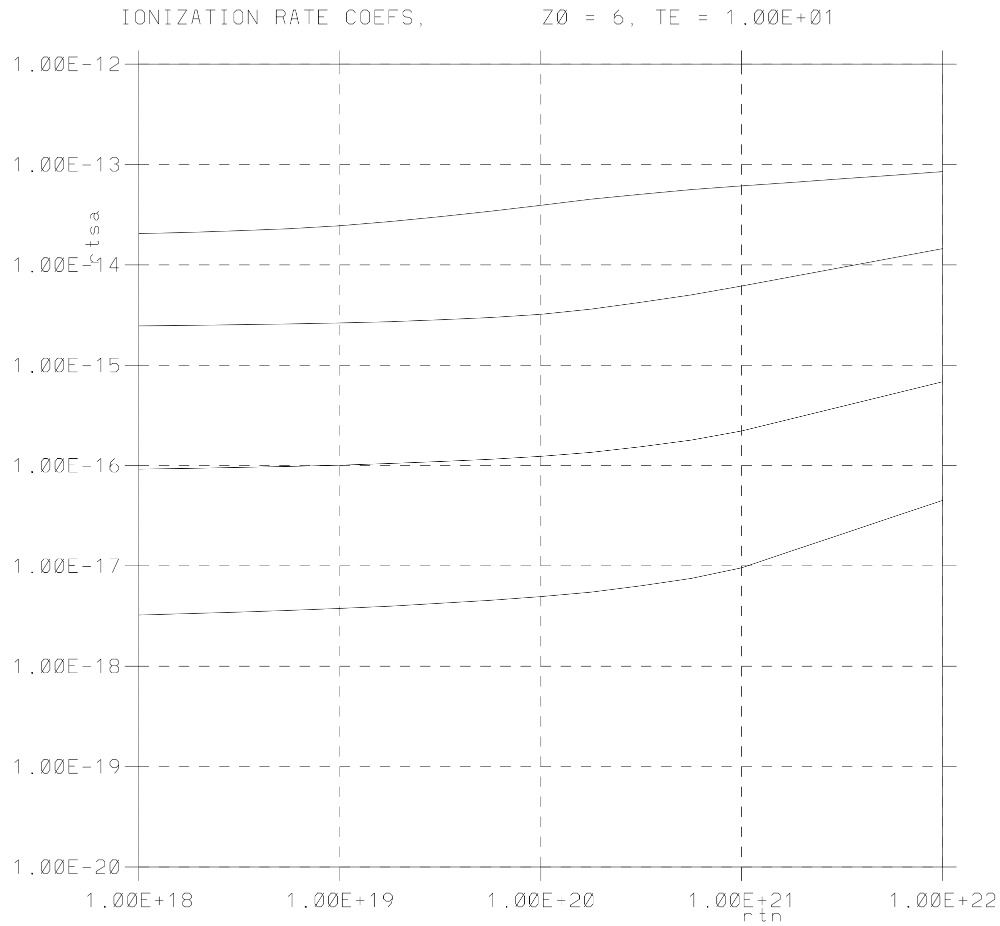


Ionization as a function of temperature for a fixed density.

STRAHL versus ADAS rates: differences in the rates for C

STRAHL

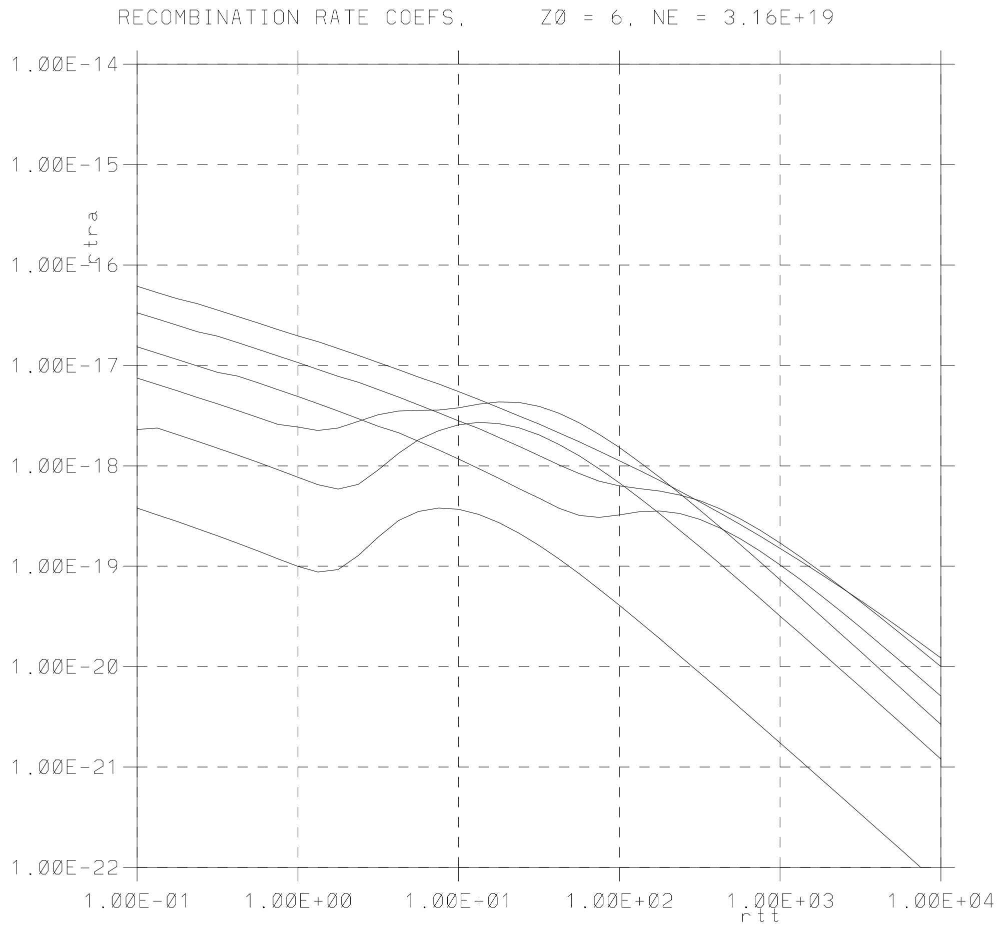
ADAS



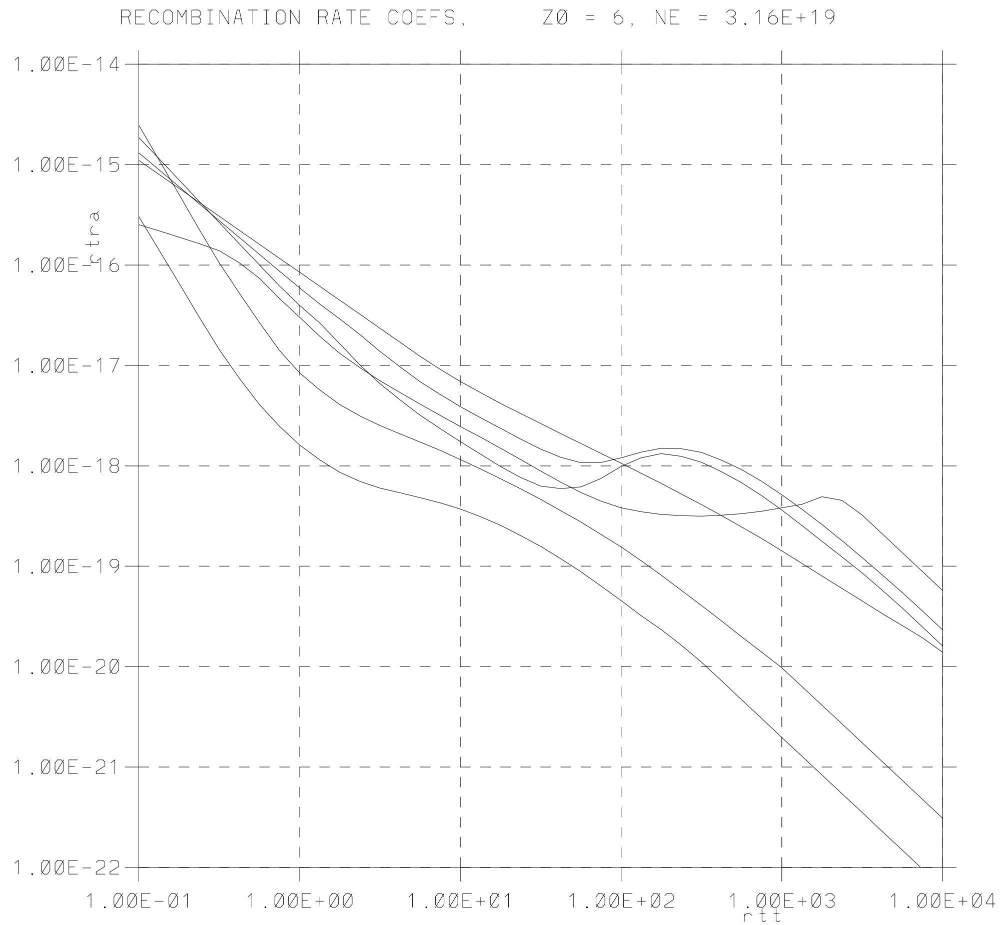
Ionization as a function of density at fixed temperature.

STRAHL versus ADAS rates: differences in the rates for C

STRAHL



ADAS

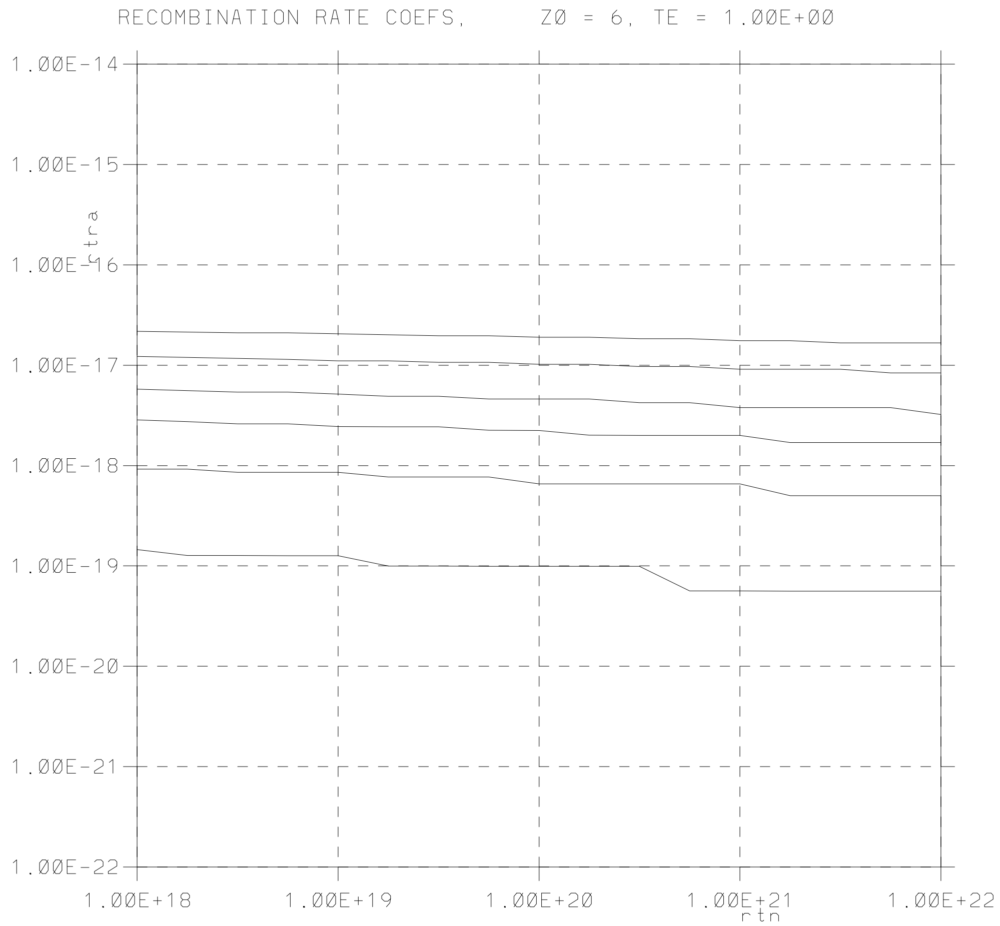


Recombination as a function of temperature for a fixed density.

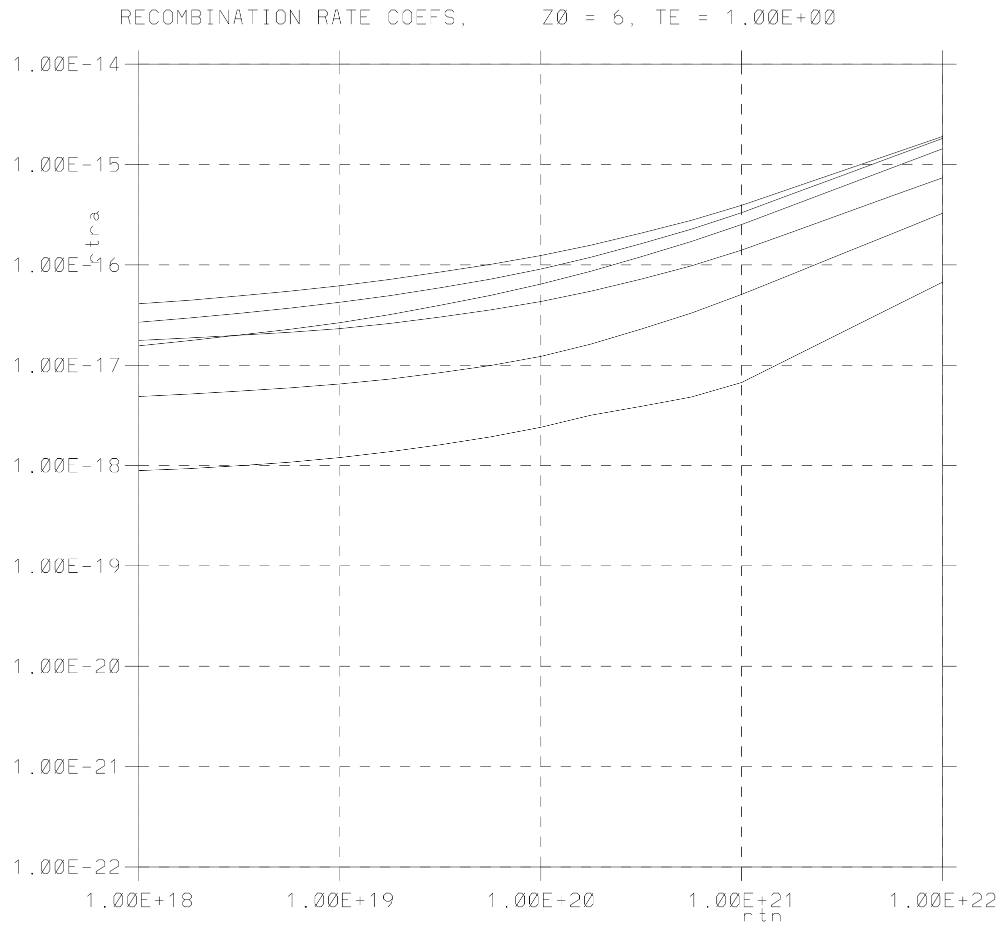
EUROPEAN FUSION DEVELOPMENT AGREEMENT

STRAHL versus ADAS rates: differences in the rates for C

STRAHL



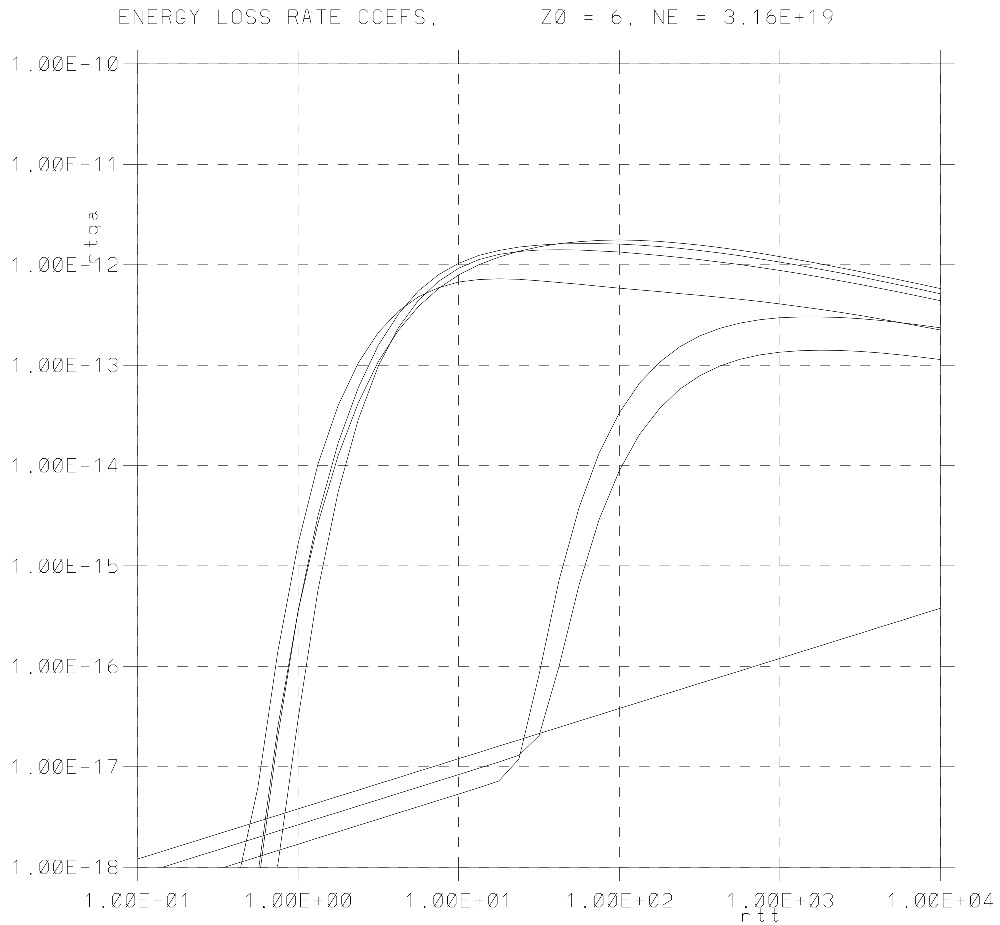
ADAS



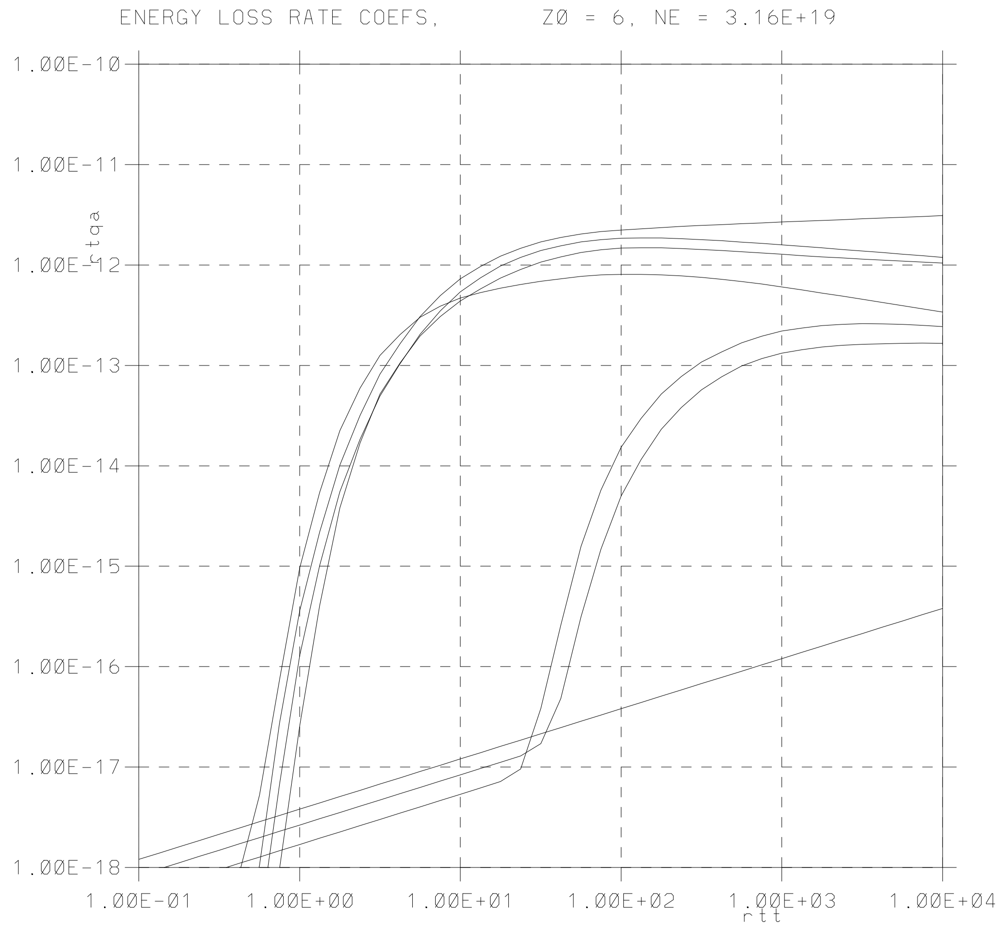
Recombination as a function of density at fixed temperature.

STRAHL versus ADAS rates: differences in the rates for C

STRAHL



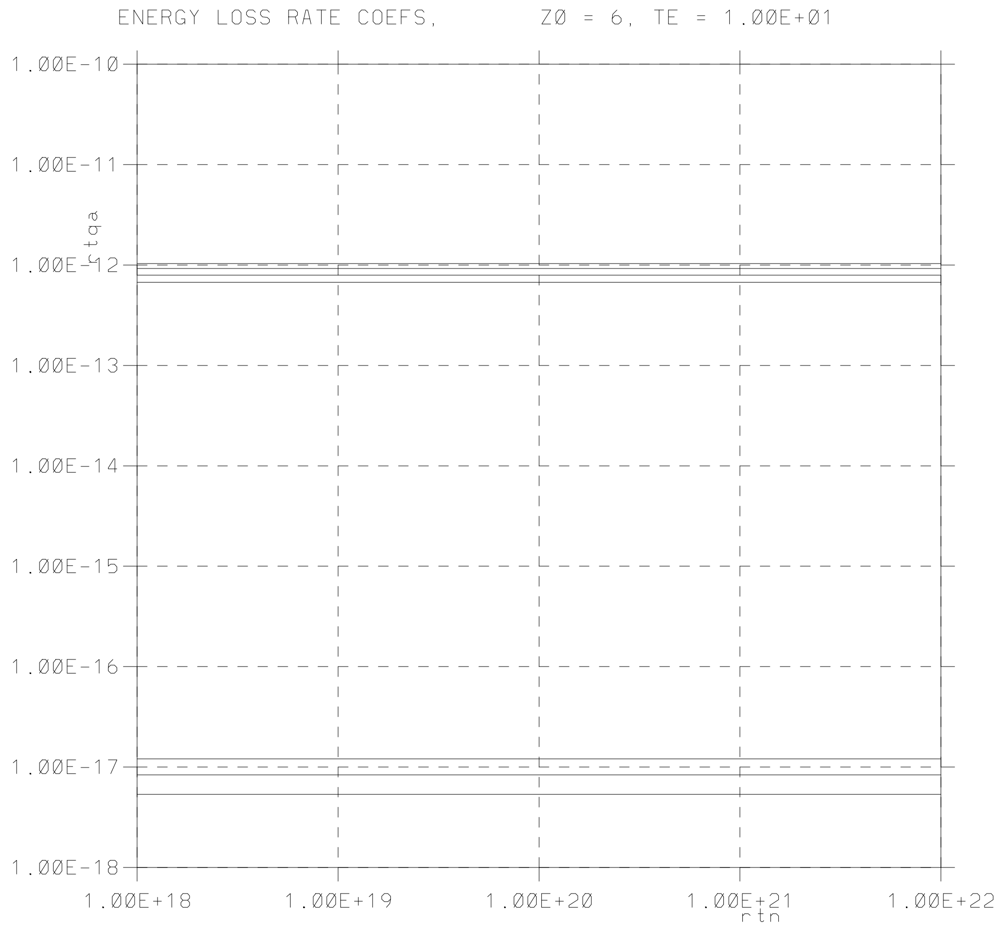
ADAS



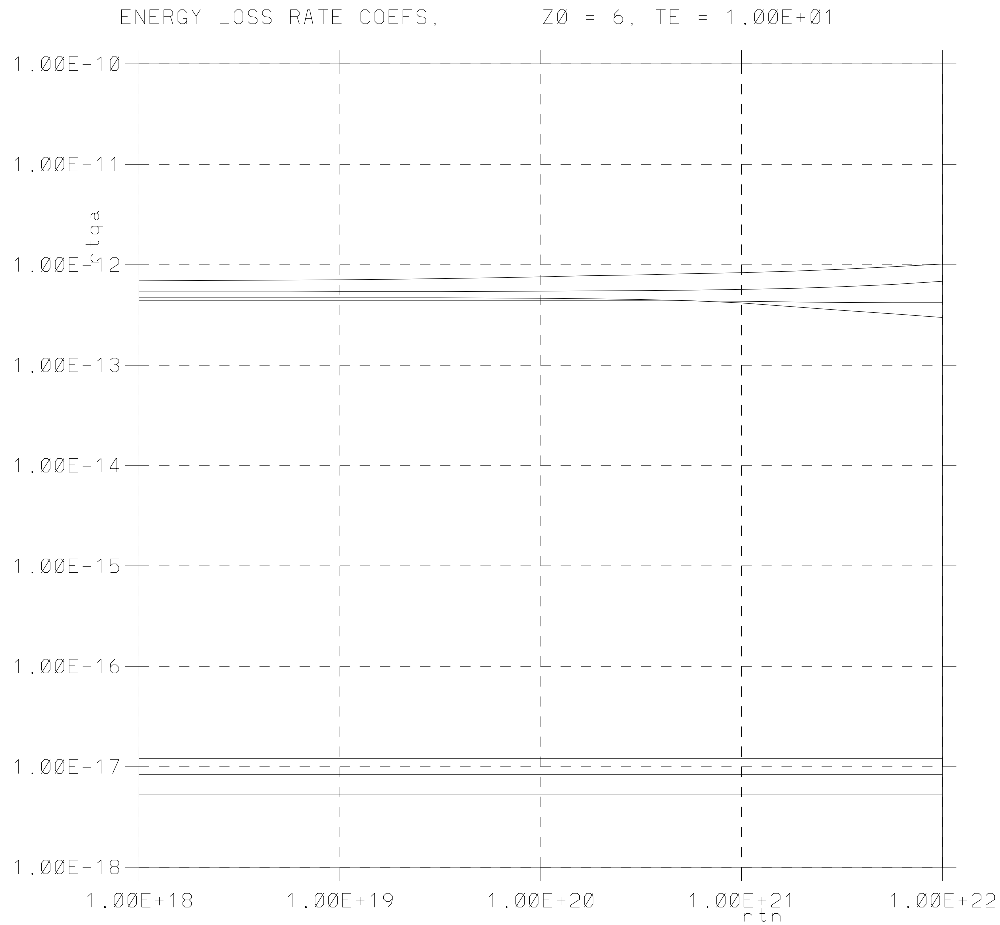
Electron cooling rates as a function of temperature for a fixed density.

STRAHL versus ADAS rates: differences in the rates for C

STRAHL



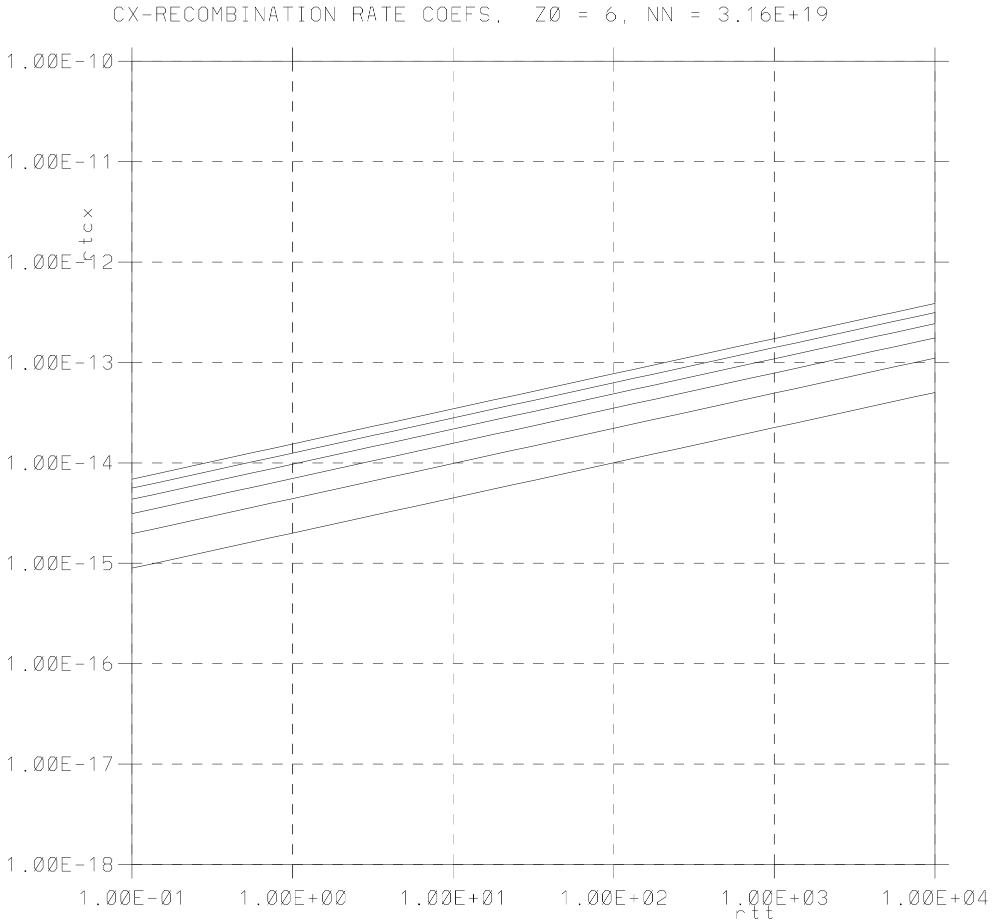
ADAS



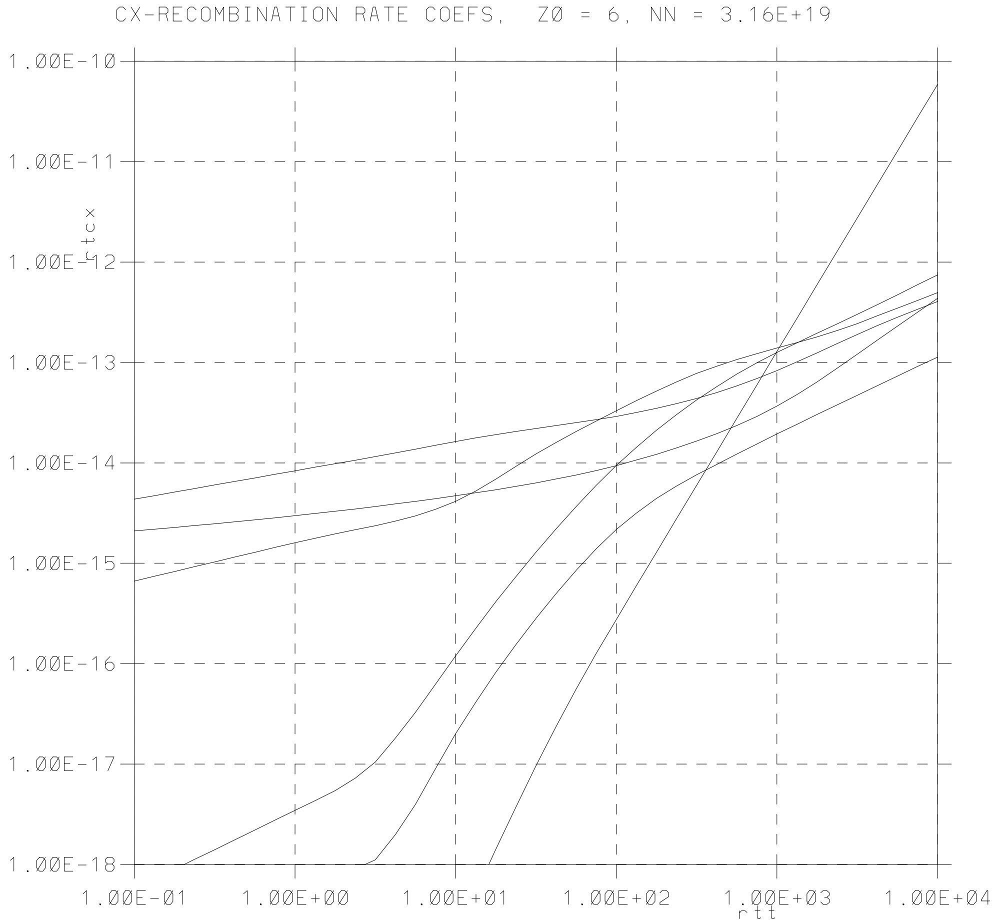
Electron cooling rates as a function of density at fixed temperature.

STRAHL versus ADAS rates: differences in the rates for C

STRAHL



ADAS

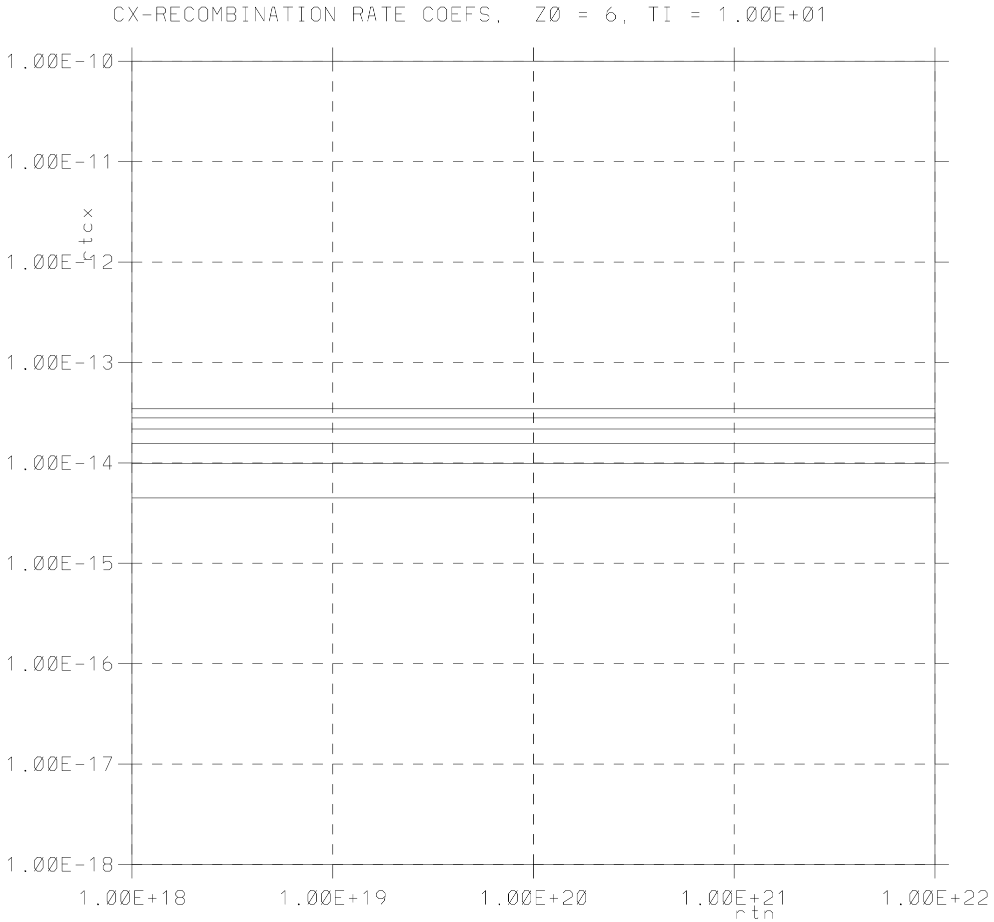


CX as a function of temperature for a fixed density.

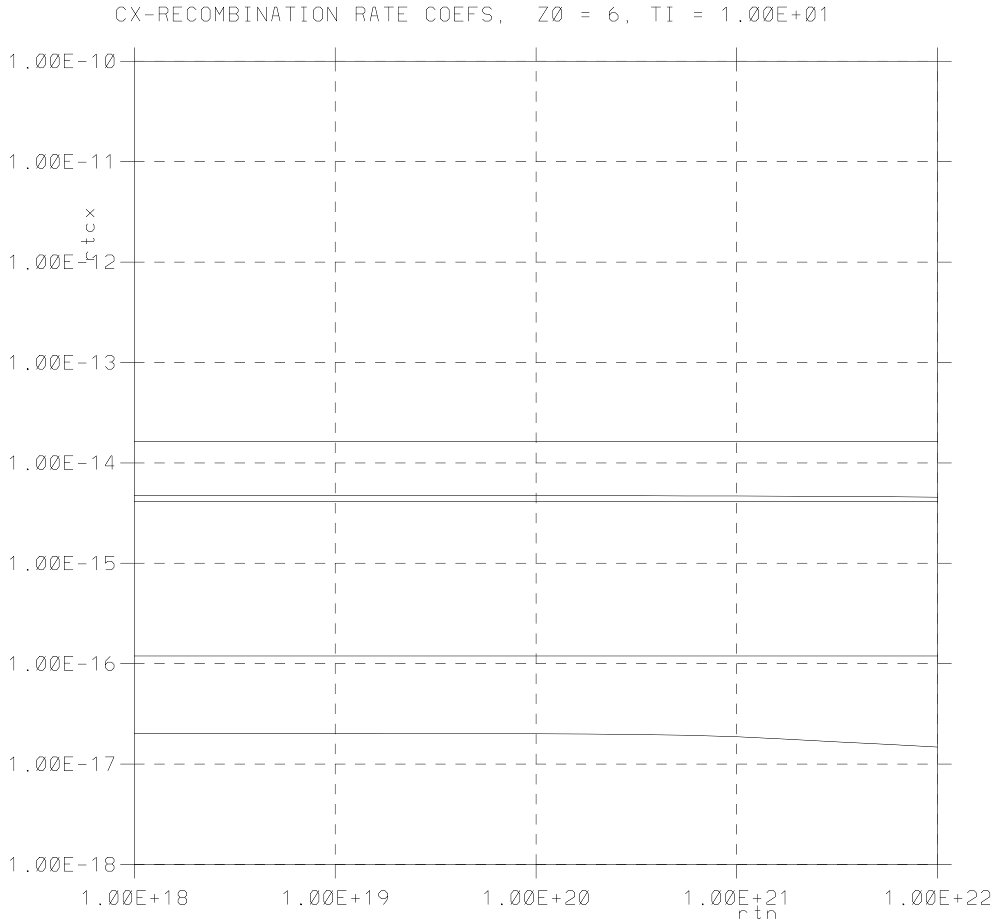
Wildly different for some charge states!

STRAHL versus ADAS rates: differences in the rates for C

STRAHL



ADAS

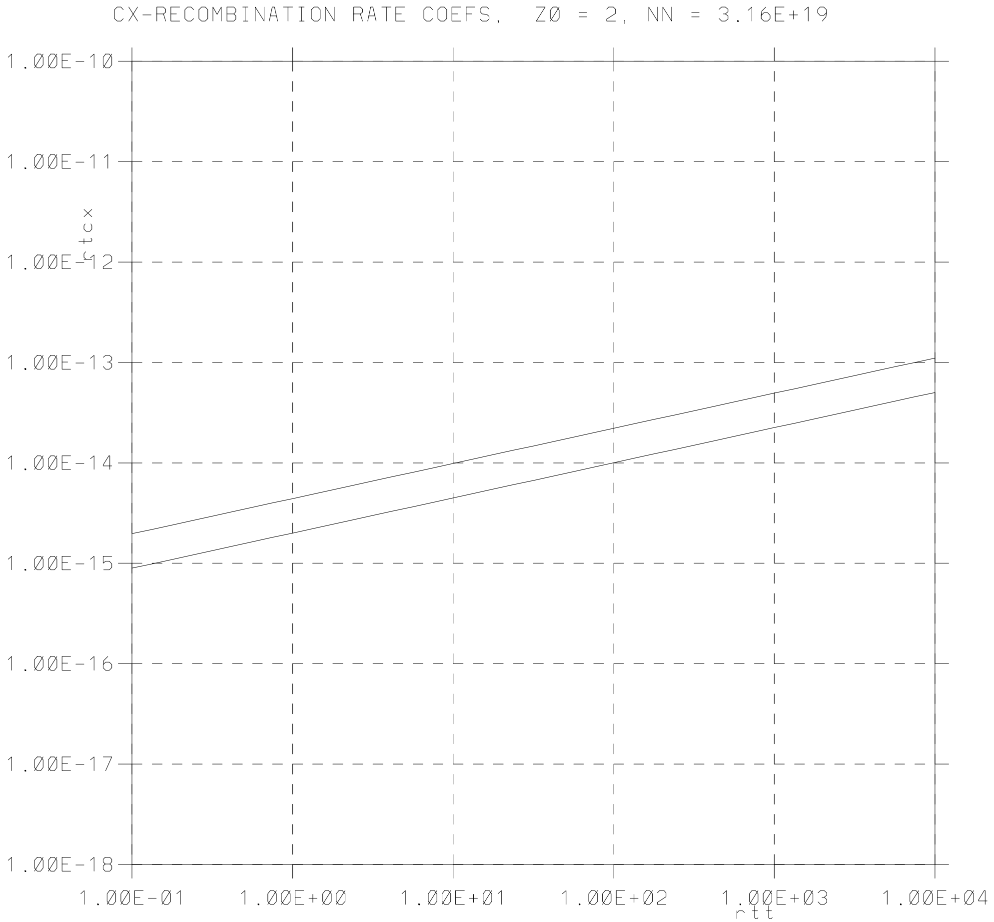


CX as a function of density at fixed temperature.

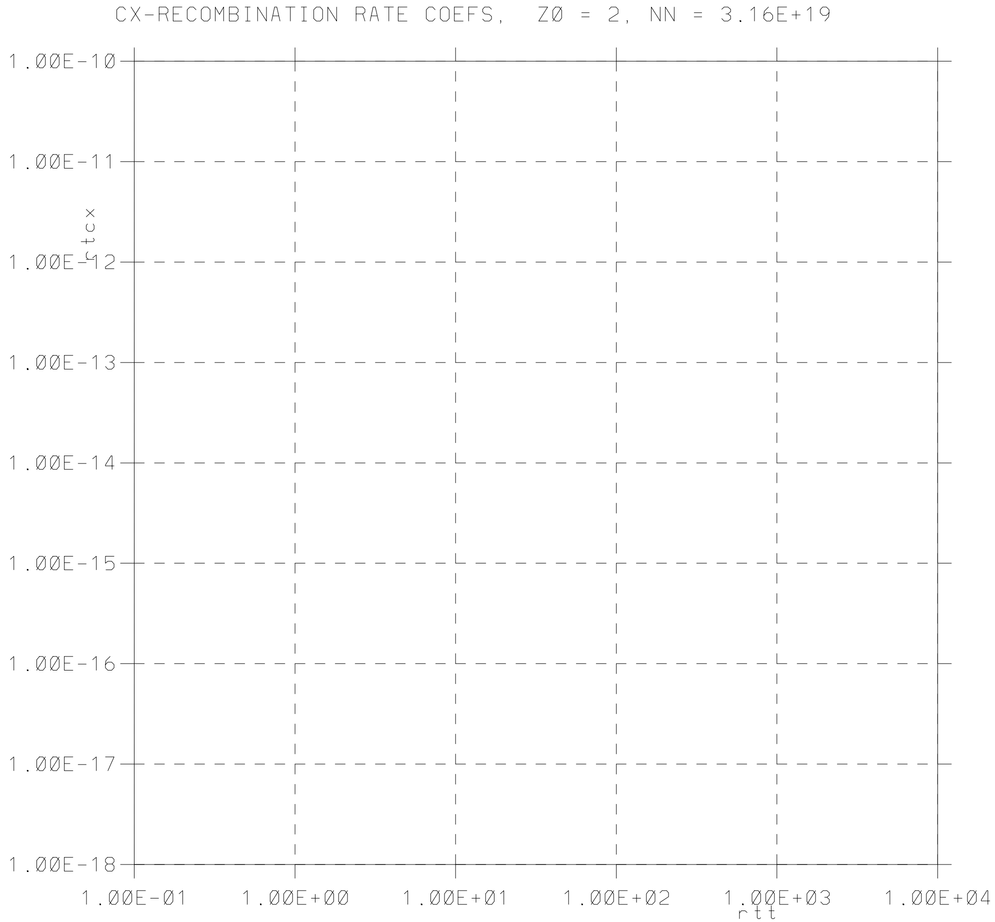
Wildly different for some charge states!

STRAHL versus ADAS rates: differences in the rates for He

STRAHL



ADAS

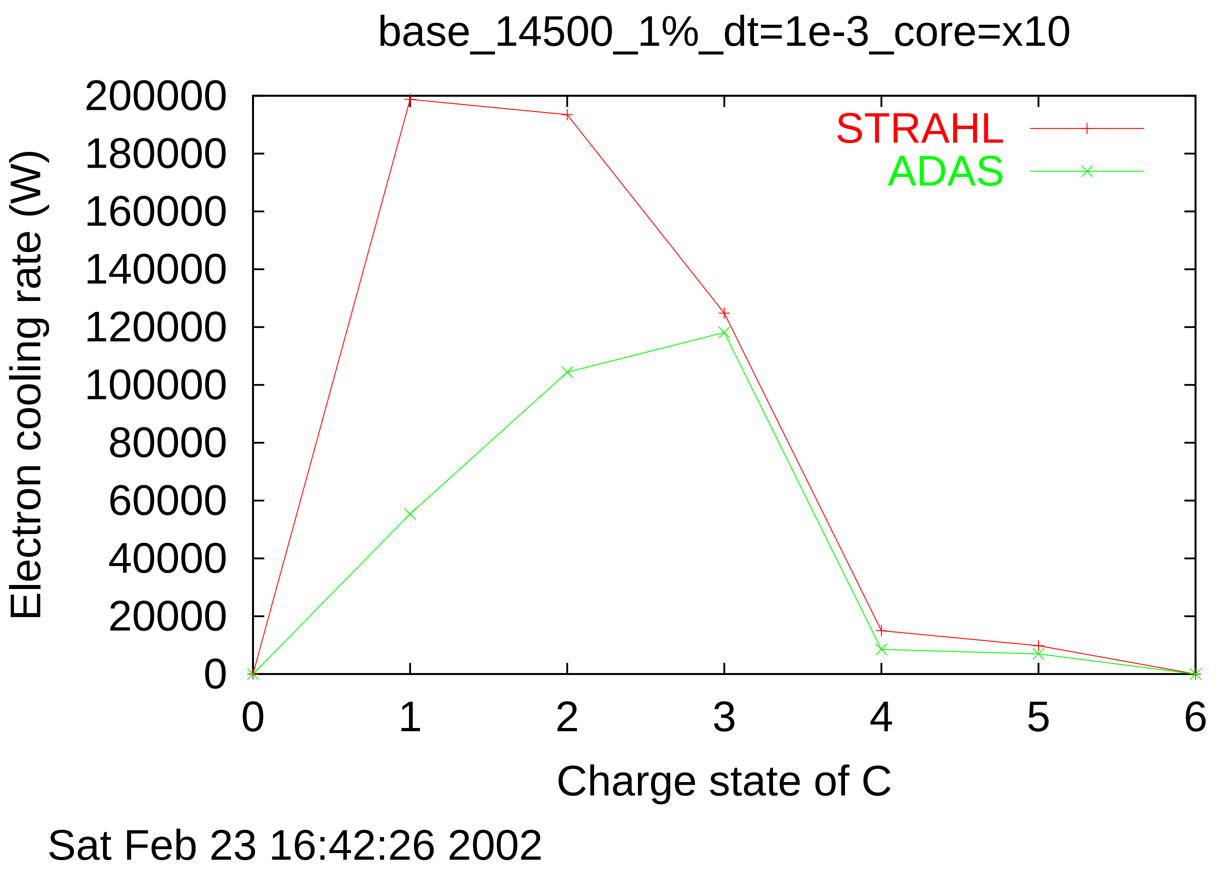


CX as a function of temperature for a fixed density.

STRAHL has Bas' fit; ADAS has nothing!

Consequences: B2.5-Eirene

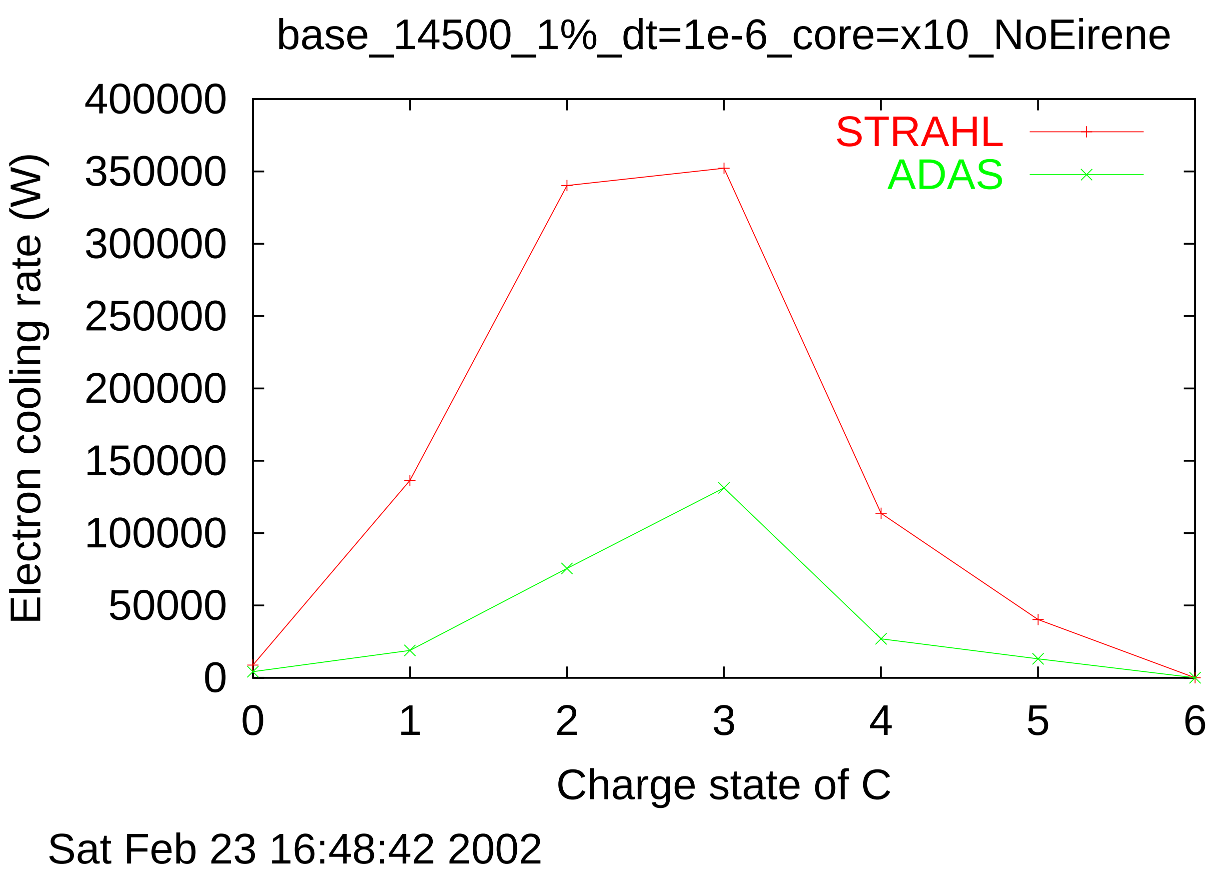
- Looking at combined run removes the CX problem
- Converged B2.5-Eirene runs — plasma conditions change



Electron cooling rate versus charge state of C for STRAHL and ADAS data.

Consequences: B2.5

- Looking at fluid neutrals run shows the combined effects (change in ionization/cooling rates as well as CX problem)
- Converged B2.5 runs — plasma conditions change



Electron cooling rate versus charge state of C for STRAHL and ADAS data.

Explanation: B2.5-Eirene

	Density	(m^{-3})		Electron cooling rate (W)		
	STRAHL	ADAS	ADAS/STRAHL	STRAHL	ADAS	ADAS/STRAHL
D^{1+}	1.54633e+19	1.51695e+19	0.981	278.41	276.133	0.991821
C^{1+}	1.31507e+16	8.06159e+15	0.613016	198781	55358.6	0.27849
C^{2+}	1.01374e+16	1.06915e+16	1.05466	193445	104362	0.539492
C^{3+}	6.91515e+15	7.79436e+15	1.12714	124832	118171	0.94664
C^{4+}	3.02678e+16	3.21165e+16	1.06108	15022.4	8533.83	0.568074
C^{5+}	2.14555e+16	2.2213e+16	1.03531	9797.04	6951.31	0.709532
C^{6+}	1.67952e+16	1.56492e+16	0.931766	10.0167	9.40951	0.939382
He^{1+}	4.33483e+14	3.98875e+14	0.920163	859.315	442.088	0.514466
He^{2+}	6.00158e+15	6.00859e+15	1.00117	0.400451	0.402124	1.00418

- Only the C^{1+} density is strongly changed
- Very strong change in the electron cooling rate for C^{1+}
- Strong changes in the cooling rate for C^{2+}, C^{4+} and He^{1+}
- Weaker change in the electron cooling rate for C^{5+}

Explanation: B2.5

	CX rate (s^{-1})		ADAS/STRAHL
D^0	-3.6903E+21	-4.9916E+20	0.135263
C^1	7.4045E+20	1.7789E+15	2.40246e-06
C^2	9.5905E+20	4.4315E+17	0.000462072
C^3	8.2139E+20	2.0695E+20	0.251951
C^4	8.6122E+20	2.1394E+20	0.248415
C^5	2.3539E+20	7.5833E+19	0.322159
C^6	5.2906E+19	1.9850E+18	0.0375194
He^1	7.5722E+18	1.2151E-46	1.60469e-65
He^2	1.2344E+19	6.3203E-47	5.12014e-66

- In addition to the changes seen for B2.5-Eirene, we see that
  - for C^1 the rate is 10^6 too high with Bas' fit
  - for C^2 the rate is 10^3 too high
  - for C^3, C^4 & C^5 the rate is a factor 3–4 too high
  - for C^6 the rate is about a factor of 30 too high

No ADAS CX data for He!

New flag for b2ar.dat

```
*tlohi           (tlo, thi; real, free format)
  1.0e-1  1.0e4
*nlohi           (nlo, nhi; real, free format)
  1.0e18  1.0e22
*numnuc          (nnuc; integer, free format)
   2
*nucspec         (nz, izlo, izhi; integers, one triple per line)
   1  0  1
   1  0  1
*flag            (either 'adpak' or 'strahl', free format)
  'adas'
*tailep          (real, free format)
  0.08
'b2ardr_fix_cx' '1'
```

b2ardr_fix_cx can take the value:

0 don't change the CX data

1 fix the hydrogen CX if the rates are 0 [default]

2 fix any CX rates if 0

3 fix H CX rates

4 fix all CX rates

- fix is done with Bas' fit formula
  - should be OK for H
  - can be wildly wrong for anything else
- (this case for two isotopes of H)

Running b2ar.exe with the ADAS option

When running with the flag set to 'adas' you should see something like

```
20:41> b2run b2ar
rm -rf b2ar.exe.dir ; mkdir b2ar.exe.dir ; cp b2ar.dat stra.dat weis.dat b2ar.exe.dir
rm -f b2ar.prt b2frates
cd b2ar.exe.dir ; time b2ar.exe ; mv b2ar.prt b2frates .. ; rm -f b2ar.dat stra.dat weis.d
 open_file: SOLPSTOP = : /space/dpc/afs/solps/solps5.0_test
 Trying ratadas.filelist
 Trying ../ratadas.filelist
 Trying /space/dpc/afs/solps/solps5.0_test/data/ratadas.filelist
 Trying ionization_potentials
 Trying ../ionization_potentials
 Trying /space/dpc/afs/solps/solps5.0_test/data/ionization_potentials
 rtsa
 rtra
 rtqa
 rtcx
 CX data 'fixed' for nuclear charge  1
 Fix based on the fit in ratsr.F by B.J.Braams
 which is probably OK for hydrogenic species
```

continued ...

Running b2ar.exe with the ADAS option, continued

continued ...

```
 Trying ratadas.filelist
 Trying ../ratadas.filelist
 Trying /space/dpc/afs/solps/solps5.0_test/data/ratadas.filelist
 Trying ionization_potentials
 Trying ../ionization_potentials
 Trying /space/dpc/afs/solps/solps5.0_test/data/ionization_potentials
 rtsa
 rtra
 rtqa
 rtcx
 CX data 'fixed' for nuclear charge  1
 Fix based on the fit in ratsr.F by B.J.Braams
 which is probably OK for hydrogenic species
 maximum use of stackw%w:        0
 maximum use of stackw%wi:       0
jwe0002i stop b2ar
3.75user 0.14system 0:06.87elapsed 56%CPU (0avgtext+0avgdata 0maxresident)k
0inputs+0outputs (877major+897minor)pagefaults 0swaps
rmdir b2ar.exe.dir
20:41>
```

Running b2ar.exe with the ADAS option, explained

From the output,

```
open_file: SOLPSTOP = : /space/dpc/afs/solps/solps5.0_test
Trying ratadas.filelist
Trying ../ratadas.filelist
Trying /space/dpc/afs/solps/solps5.0_test/data/ratadas.filelist
```

the program looks for the ADAS data by trying to read `ratadas.filelist`,

1. in the current directory (b2ar.exe.dir)
2. in the directory in which you started the `b2run b2ar`
3. in `${SOLPSTOP}/data/ratadas.filelist`

ratadas.filelist

This file contains

```
'/afs/ipp-garching.mpg.de/u/adas/adas'
```

giving the top level directory to be used for ADAS (this for IPP-Garching, it is different for JET)

```
18
```

the number of species in the file (this will need to be changed if we ever need anything higher than Ar)

```
'96  ' '96  ' '96  ' '96  ' '96  ' '96  ' '96  ' 'h '
```

which give the "ADAS" year, respectively, for

- recombination
- ionization
- charge exchange
- bremsstrahlung (not used)
- line radiation (which is converted into an electron cooling rate)
- dummy
- dummy
- species

ratadas.filelist, continued

the rest of the species

```
'96  ' '96  ' '96  ' '96  ' '96  ' '96  ' '96  ' 'he'
'none' 'none' 'none' 'none' 'none' 'none' 'none' 'li'
'93  ' '93  ' '93  ' '93  ' '93  ' '93  ' '93  ' 'be'
'89  ' '89  ' '89  ' '89  ' '89  ' '89  ' '89  ' 'b '
'96  ' '96  ' '96  ' '96  ' '96  ' '96  ' '96  ' 'c '
'96  ' '96  ' '96  ' '96  ' '96  ' '96  ' '96  ' 'n '
'96  ' '96  ' '96  ' '96  ' '96  ' '96  ' '96  ' 'o '
'89  ' '89  ' '89  ' '89  ' '89  ' '89  ' '89  ' 'f '
'89  ' '89  ' '89  ' '89  ' '89  ' '89  ' '89  ' 'ne'
'none' 'none' 'none' 'none' 'none' 'none' 'none' 'na'
'none' 'none' 'none' 'none' 'none' 'none' 'none' 'mg'
'none' 'none' 'none' 'none' 'none' 'none' 'none' 'al'
'89  ' '89  ' '89  ' '89  ' '89  ' '89  ' '89  ' 'si'
'none' 'none' 'none' 'none' 'none' 'none' 'none' 'p '
'89  ' '89  ' '89  ' '89  ' '89  ' '89  ' '89  ' 's '
'89  ' '89  ' '89  ' '89  ' '89  ' '89  ' '89  ' 'cl'
'89  ' '89  ' '89  ' '89  ' '89  ' '89  ' '89  ' 'ar'
```

ratadas.filelist, continued

and then some comments

```
  Change log:
          2000.07.10      DPC     Changed h, he, c, n, o from 89 to 96
                                  Changed be to 93
                                  Recommended by Behringer and Horton
```

The conversion of the line radiation to electron cooling rate (which is what is used by B2.5) is done by adding

- a Bremsstrahlung contribution (from an analytic formula)
- the line radiation
- the ionization rate times the ionization potential [check that this is done properly]
- the contribution from recombination is currently neglected